Draft The Hidden Chaos from Randomness

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Abstract

Traditionally, chaotic signals are studied separately from those of stochastic origin. This paper shows that by applying various signal decomposition methods in the time domain to sequences of random numbers such as pseudo-random numbers, signals emerge in these decompositions that verify the properties of those corresponding to chaotic processes, although it is established that there are proposals to generate random sequences from chaotic signals, the present study goes in a somewhat opposite direction, i.e., it presents the presence of chaos in the decomposition of sequences of both random and pseudo-random signals.

1 Introduction

In current data and signal analysis, the study of random and pseudo-random signals has been carried out using tools and pursuing objectives that differentiate them from studies of chaotic signals. They can be conceptualized as two almost parallel lines of research, which in few studies have points of interest that intersect these nodes of analysis. Even those that, based on certain behaviors in monitoring the evolution of a given system, change characteristics. There are many ways to analyze the link between the random or pseudo-random signals and chaotic ones. The links between the random or pseudo-random process and chaos was explored with a interesting point of view by [Crutchfield, 2012] for example . The use of chaotic signals to construct random sequences is a field of known research in the literature, i.e. see the review of Naik and Singh [Naik and Singh, 2024]. In other subject, the transition from random to chaos was explored as has been reported by [Yu et al., 1990].

From here on, when stochastic signals are mentioned, both random and pseudo-random signals will be included.

The methods used in the decomposition of stochastic signals are Empirical Mode Decomposition (EMD) [Wang et al., 2010] [Huang et al., 1998] [Rilling et al., 2003], Variational Mode Decomposition (VMD) [Boyd et al., 2011] [Dragomiretskiy and Zosso, 2013] [Moody and Mark, 2001], Empirical Wavelet Decomposition (EWD) [Gilles, 2013], and Multiresolution Analysis (MRA) [Percival and Walden, 2000].

2 Methodology

2.1 Data Sources

Random signals obtained from [Haahr, 2025] originated from atmospheric noise. Pseudo-random numbers generated by computer algorithms white noise , Gaussian white noise [Zhivomirov, 2018]

2.2 Decomposition signal schemes

EMD VMD EWD AMR

2.3 Statistical Test

for the statistic test was employed the well known the runs test [Gibbons and Chakraborti, 2014] [Smeeton et al., 2025] for testing the randomness of the origninal aleatory and pseudoaleatory signals studied and for the analysis of the components after de decomposition en each case.

2.4 Chaotic Test

[BenSaïda, 2015]

2.5 Entropy Complexity Plane

[Amigó et al., 2006], [Amigó et al., 2008], [Amigó, 2010], [Rosso et al., 2012], [Rosso et al., 2013] [Weygand and Kivelson, 2019] and [Maggs and Morales, 2013]

3 Results

3.1 Chaotic Signals Obtained with Empirical Mode Decomposition

Just one example: fractional random singal obtained from atmospheric noise, 10000 data points.

Table I: *Characteristics of the data series analyzed (signal length 10.000 data points)*

Type	LE	CHT	ST	H	C	Mean	Var	Std
Gaussian White Noise	-0.0065	1	0	0.9943	0.0138	0.0051	0.9678	0.9838
White Noise	-0.0029	1	0	0.9944	0.0136	0.004	0.9947	0.9974
Atmospheric Noise	-0.0349	1	0	0.9940	0.0145	0.4997	0.0827	0.2876

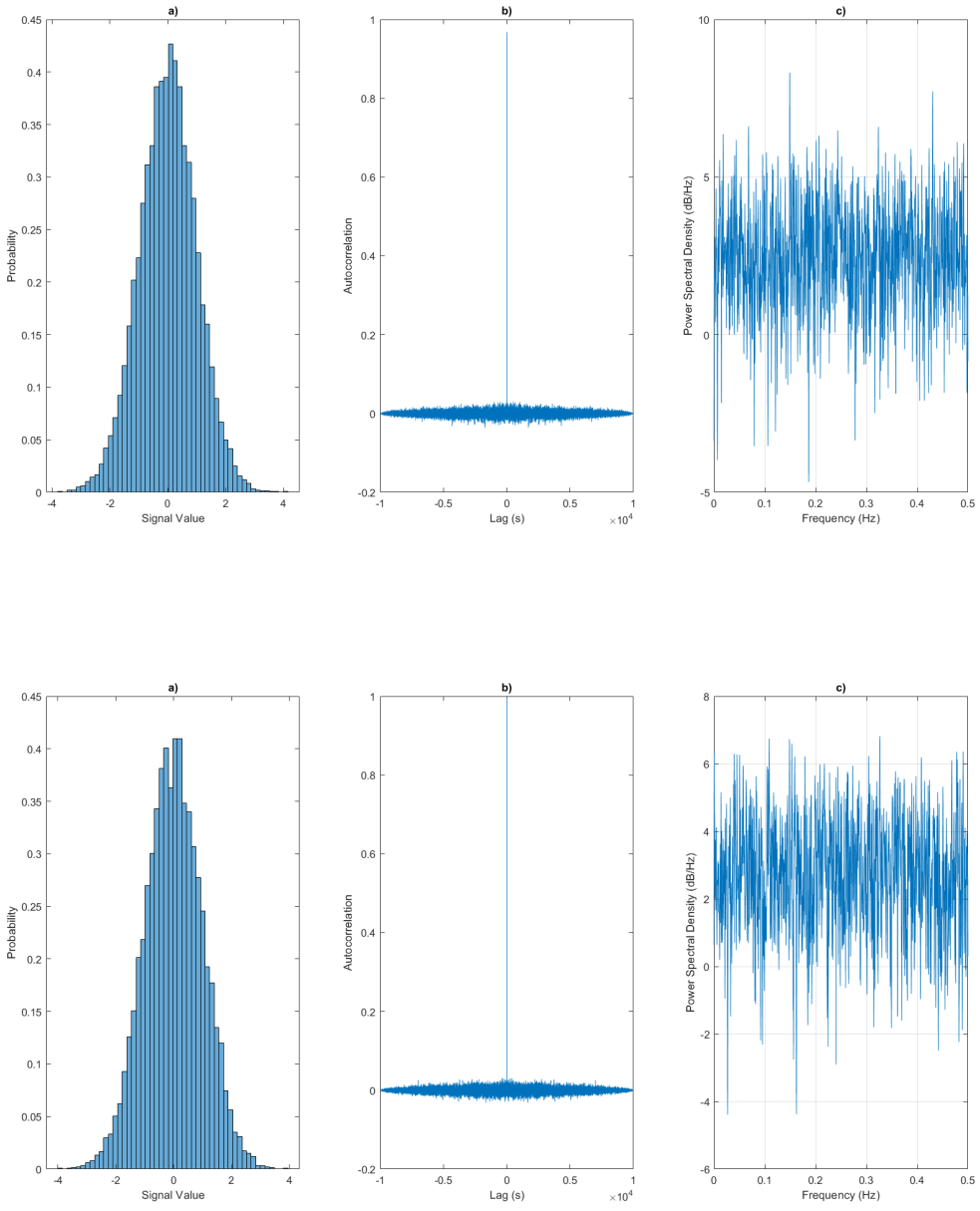


Figure 1: *White Noise: a) Histogram, b) Autocorrrelation, c) Power Spectral Density*

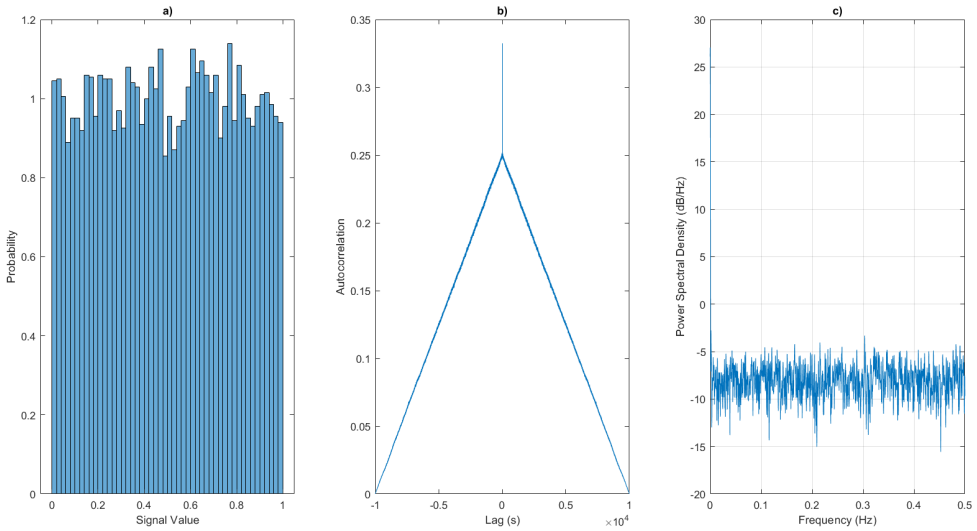


Figure 2: *Random Signal: a) Histogram, b) Autocorrrelation, c) Power Spectral Density*

4 Results

4.1 Chaotic Signals Obtained Using Empirical Mode Decomposition

Table II: *Results for atmospheric noise 10 leveles of EMD decomposition. LE means Lyapunov Exponent, CHT chaos test, and ST statistical RUNS test*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.2325	0.3746	0.5937	0.5743	0.4352	0.3380	0.3268	0.3090	0.2914	-7.4981e-6
CHT	1	1	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	1	1	1	0
H	0.9545	0.6297	0.3830	0.2461	0.1757	0.1389	0.1198	0.1124	0.1075	0.1065
C	0.1066	0.4383	0.3392	0.2282	0.1668	0.1342	0.1170	0.1105	0.1064	0.1056

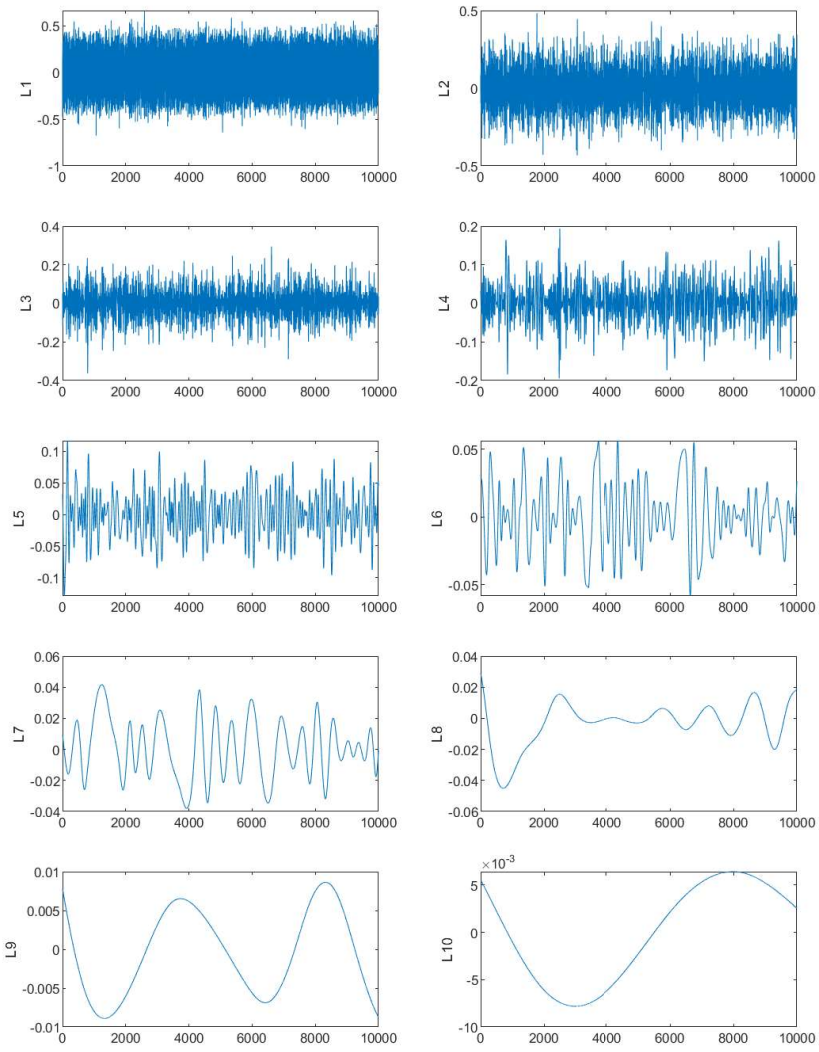


Figure 3: *EMD of the Atmospheric Noise 10 Levels*

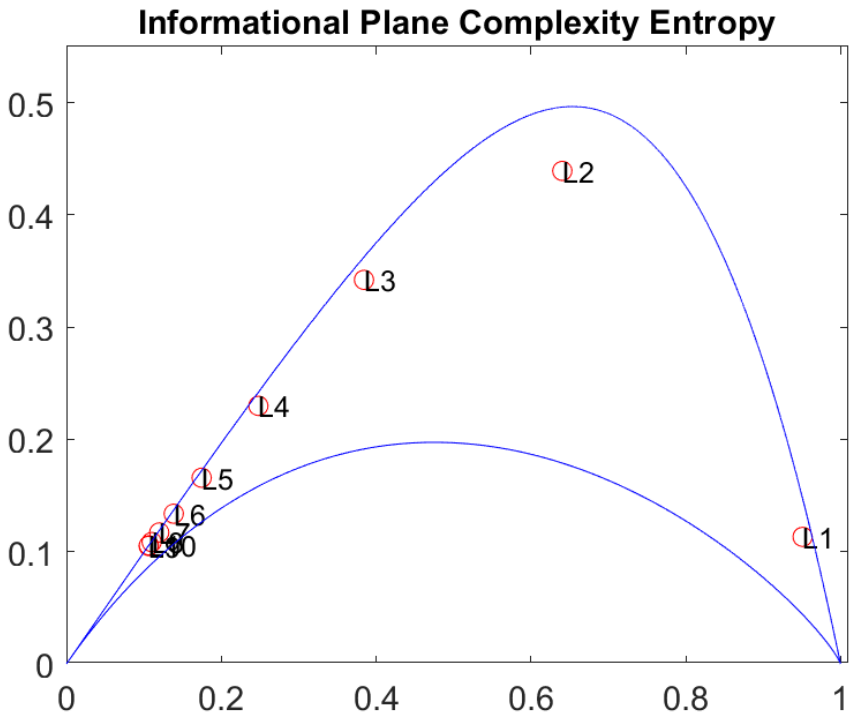


Figure 4: *Entropy Complexity Plane of the EMD of the Atmospheric Noise*

Table III: *Results for Gaussian white noise 10 leveles of EMD decomposition. LE means Lyapunov Exponent, CHT chaos test, and ST statistical RUNS test*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.2341	0.3615	0.5875	0.5714	0.4325	0.3447	0.3198	0.3383	0.3008	0.0140
CHT	1	1	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	1	1	1	1
H	0.9545	0.6420	0.3812	0.2439	0.1711	0.1356	0.1202	0.1089	0.1022	0.1059
C	0.1044	0.4373	0.3385	0.2267	0.1630	0.1312	0.1374	0.1072	0.1012	0.1051

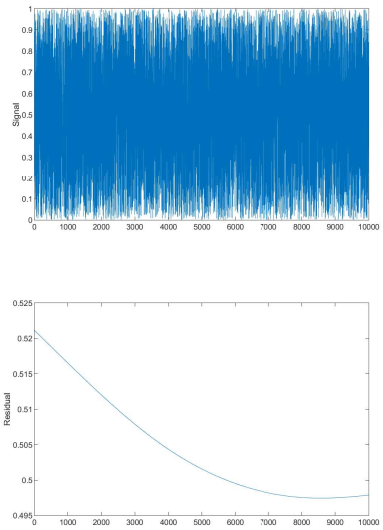


Figure 5: *Signal and Residual of EMD Atmospheric Noise 10 Levels*

Table IV: *Results for white noise 10 leveles of EMD decomposition. LE means Lyapunov Exponent, CHT chaos test, and ST statistical RUNS test*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.2333	0.3660	0.5932	0.5847	0.4378	0.3384	0.3223	0.2955	0.1448	0.0448
CHT	1	1	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	1	1	1	1
H	0.9525	0.6481	0.3847	0.2461	0.1724	0.1365	0.1190	0.1112	0.1068	0.1059
C	0.1093	0.4365	0.3403	0.2284	0.1640	0.1321	0.1163	0.1095	0.1058	0.1051

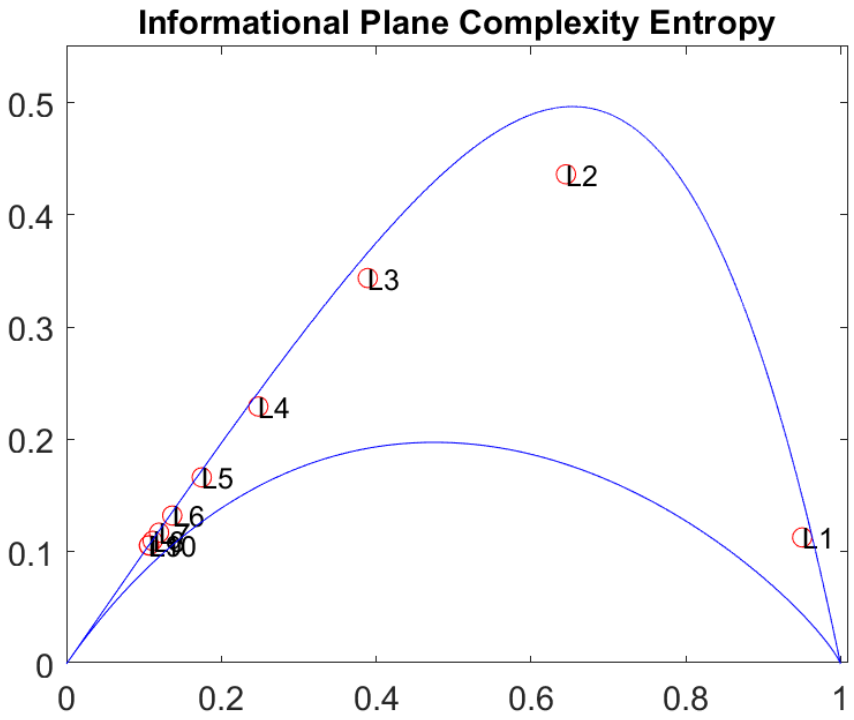


Figure 6: *Entropy Complexity Plane of the EMD of the Gaussian White Noise*

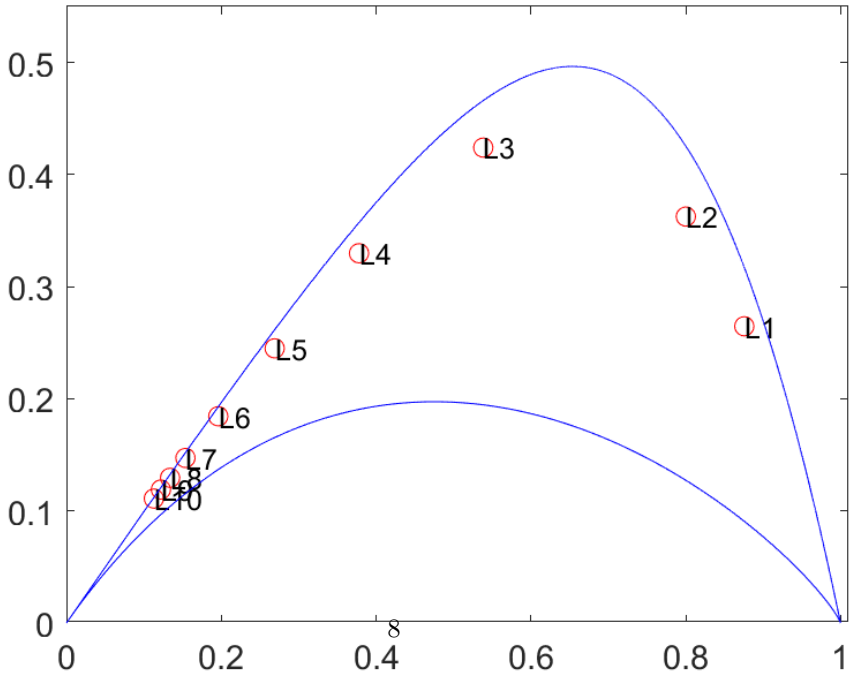


Figure 7: *Entropy Complexity Plane of the EMD of the White Noise*

4.2 Chaotic Signals Obtained with Variational Mode Decomposition

Table V: *Results for atmospheric noise with VMD decomposition*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.5042	0.3562	0.3517	0.4744	0.5060					
CHT	0	1	0	0	0					
ST	1	1	1	1	0					
H	0.6556	0.7160	0.7033	0.5195	0.4806					
C	0.4432	0.4017	0.4219	0.4113	0.3388					

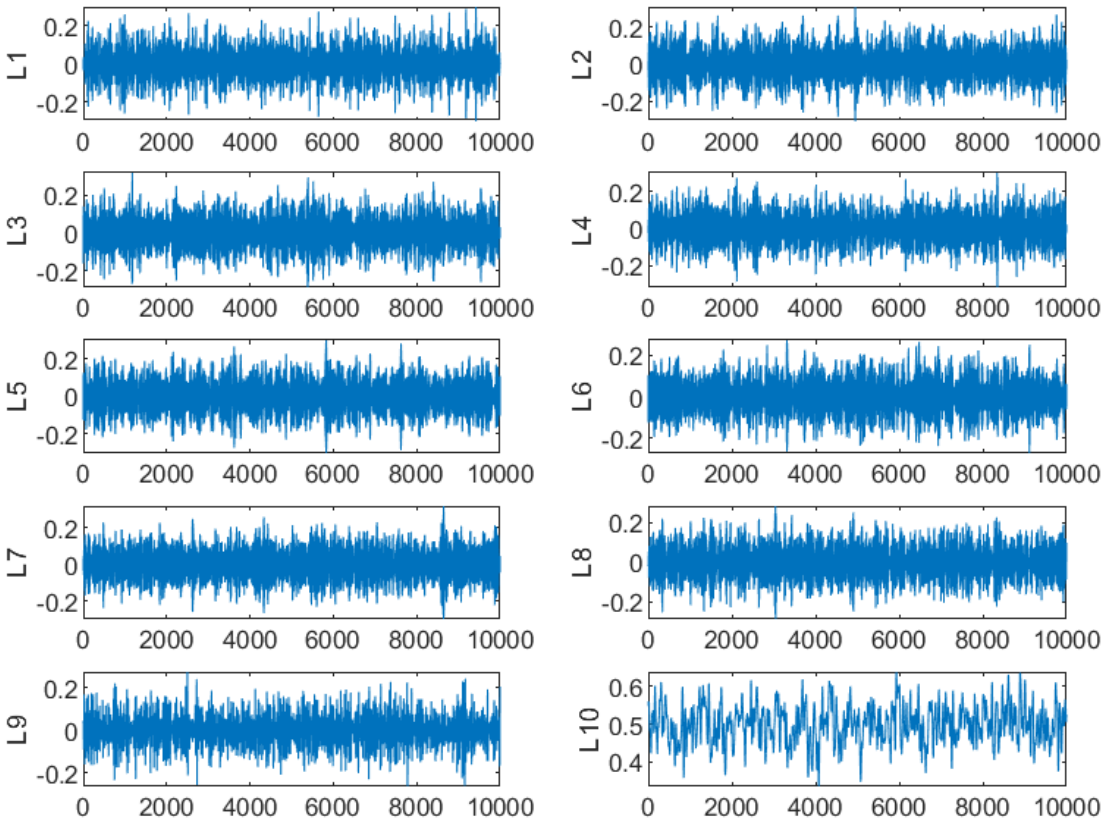


Figure 8: *Atmospheric Noise VMD 10 Levels*

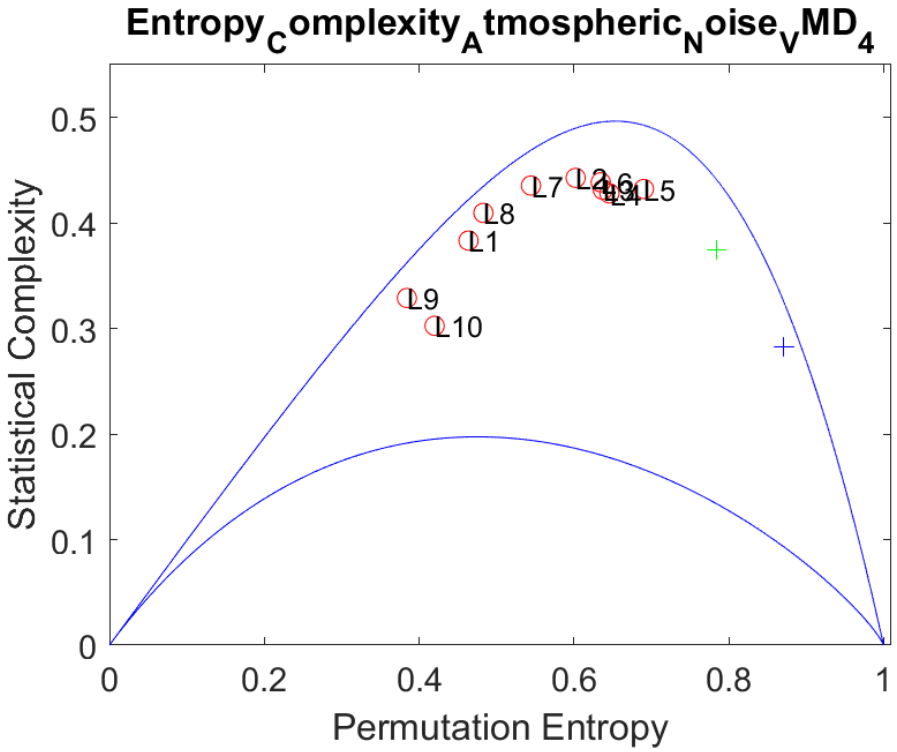


Figure 9: *Entropy Complexity Plane Atmospheric Noise VMD 10 Levels*

Table VI: *Results for Gaussian White noise with VMD decomposition*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9 L10	
LE	0.5627	0.3996	0.3505	0.3555	0.4869					
CHT	1	1	0	1	1					
ST	1	1	1	1	1					
H	0.5183	0.6813	0.7065	0.6713	0.5060					
C	0.4048	0.4371	0.4094	0.4349	0.4131					

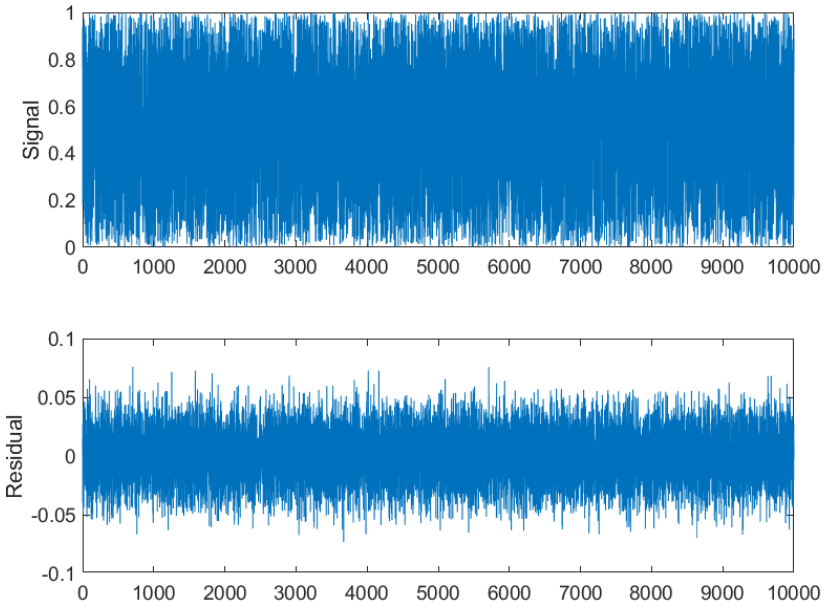


Figure 10: *Atmospheric Noise Signal and Residual*

Table VII: *Results for White noise with VMD decomposition*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.5565	0.4097	0.3490	0.3592	0.5200					
CHT	1	0	0	1	0					
ST	1	1	1	1	1					
H	0.5115	0.6709	0.7081	0.6507	0.4928					
C	0.4007	0.4412	0.4086	0.4391	0.3994					

4.3 Chaotic Signals Obtained Using Empirical Wavelet Transform

Table VIII: *Results for atmospheric noise with EWT decomposition*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.2482	0.2538	0.0598	0.3737	0.3833	0.2025	0.0936	0.2730	0.3940	0.4809
CHT	1	0	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	1	1	1	1
H	0.8910	0.4753	0.4441	0.6544	0.5576	0.4417	0.4394	0.4471	0.4793	0.4592
C	0.2457	0.4102	0.4017	0.4380	43.08	0.3982	0.4002	0.4021	0.4166	0.4010

Table IX: *Results for Gaussian white noise with EWT decomposition*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.5926	0.5931	0.4355	0.3819	0.3662	0.2708	0.3644	0.3738	0.3577	0.3535
CHT	0	0	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	0	1	1	1
H	0.3744	0.5247	0.6455	0.6782	0.5930	0.4603	0.7235	0.6351	0.4901	0.4775
C	0.3334	0.4244	0.4481	0.4258	0.4358	0.4049	0.4155	0.4411	0.4167	0.4115

Table X: *Results for white noise with EWT decomposition*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.4447	0.3801	0.3421	0.3166	0.4061	0.5910	0.5352	0.3119	0.3426	0.3607
CHT	0	0	0	0	0	0	0	0	0	0
St	1	1	1	1	1	1	1	1	1	1
H	0.2915	0.3317	0.3525	0.3600	0.3756	0.4579	0.4957	0.4516	0.4617	0.4706
C	0.2712	0.3084	0.3267	0.3344	0.3454	0.3975	0.4187	0.4032	0.4070	0.4062

4.4 Chaotic Signals Obtanined with Multi Resolution Analysis

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
LE	0.2775	0.2962	0.4355	0.5881	0.5777	0.4619	0.3551	0.3379	0.3250	0.3161
CHT	1	1	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	1	1	1	1
H	0.8756	0.8002	0.5384	0.3780	0.2690	0.1961	0.1537	0.1339	0.1220	0.1130
C	0.2647	0.3626	0.4241	0.3298	0.2450	0.1844	0.1472	0.1295	0.1190	0.1110

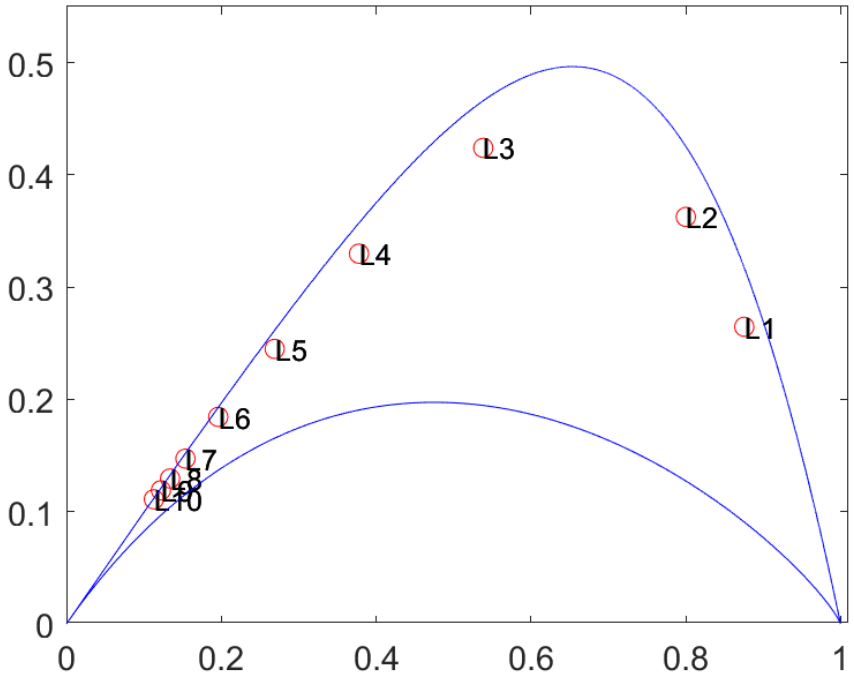


Figure 11: *Entropy Complexity Plane of the Atmospheric Noise Decomposed using AMR*

Table XI: *Decomposition of Gaussian White Noise Using AMR*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	:10
LE										
CHT	1	0	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	1	1	1	1
H										
C										

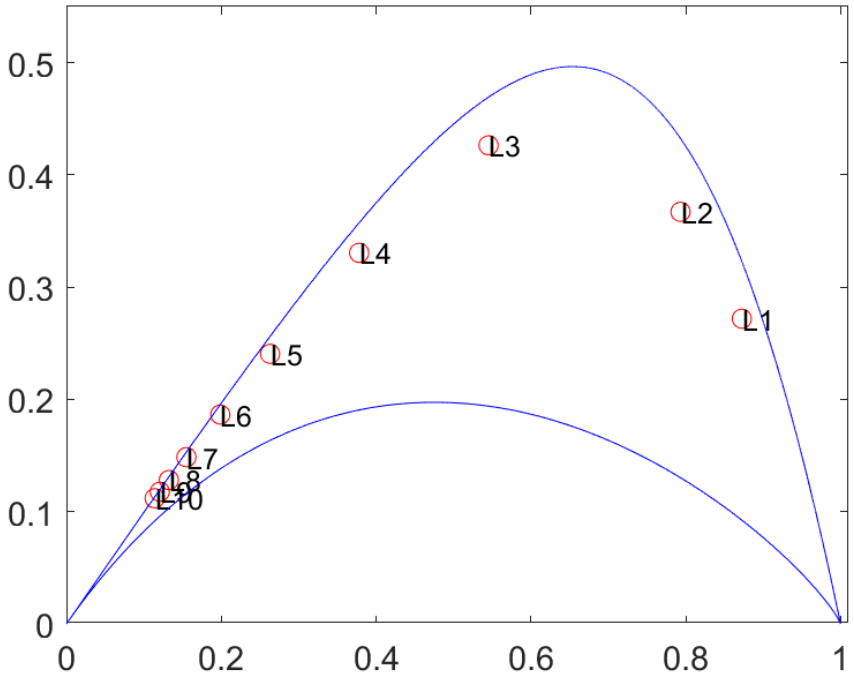


Figure 12: *Entropy Complexity Plane Gaussian White Noise using AMR*

Table XII: *Decomposition of White Noise Using AMR*

Indicator	L1	L2	L3	L4	L5	L6	L7	L8	L9	:10
LE										
CHT	1	0	0	0	0	0	0	0	0	0
ST	1	1	1	1	1	1	1	1	1	1
H										
C										

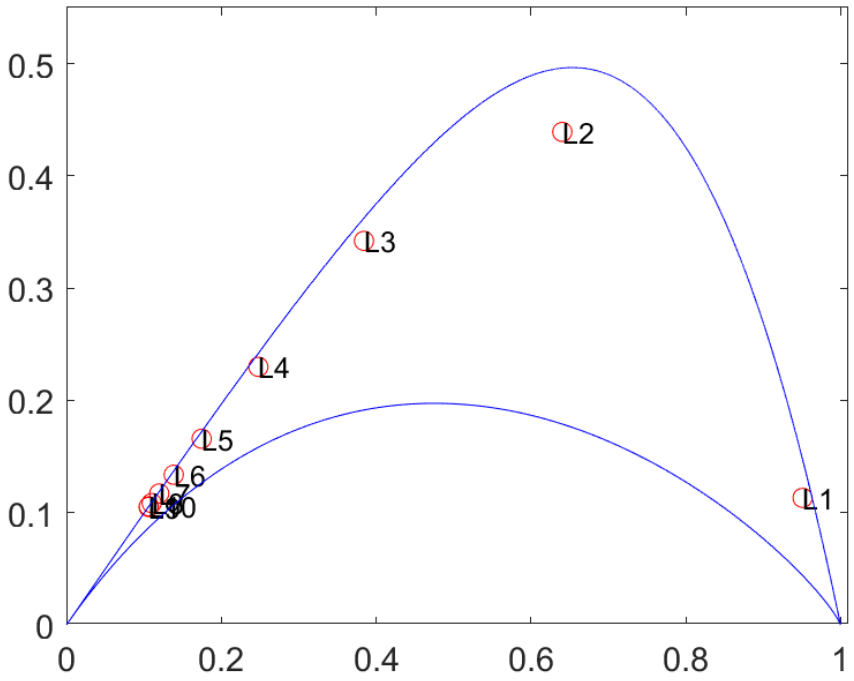


Figure 13: *Decomposition of White Noise using AMR*

5 Results Analysis

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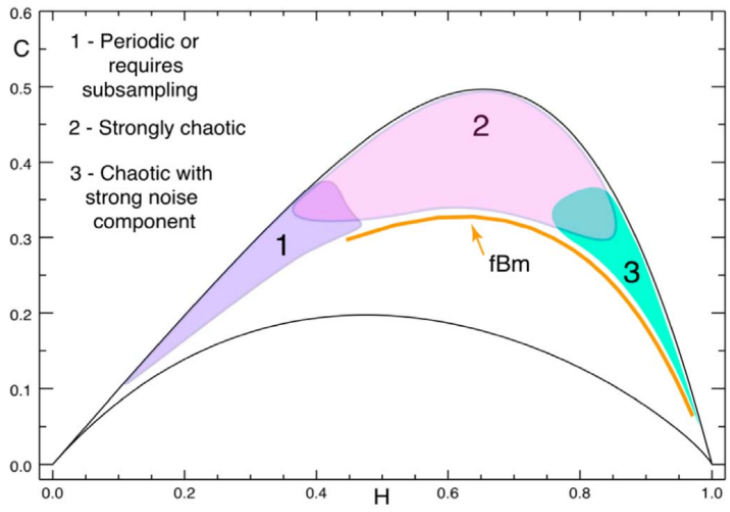


Figure 14: *HXC Zonas of Characteristic Behaviour*

6 Conclusions

In all cases analyzed in this research, both for pseudo-random and random signals, and regardless of the type of algorithm, it was found that one of the components of the primary stochastic signals was chaotic in nature. This was confirmed by statistical tests and classical tests. In addition to the chaotic component, in some cases other random components were found, as well as some that were neither random nor chaotic, but deterministic. This result can be interpreted to assume that what is called randomness is made up of a chaotic component, and therefore sensitive to initial conditions, and deterministic components and, in some cases, other random components. It can therefore be thought that the results of tossing a coin are composed of a chaotic component, sensitive to the initial conditions of how the coin is tossed into the air, and others of a deterministic nature given by the laws of Newtonian dynamics, and in some cases some extra random component such as the action of a current of air on the coin when it is in the air. In other order topic, the results presented could be used as a new method to generate chaotic sequences and generate new attractors The results presented in this report could be pointing to a new way of interpreting complexity in terms of random processes being made up of chaotic components, combined with some random elements of lesser variability, which implies lower entropy and, in some cases, greater statistical complexity, and other deterministic elements of an oscillatory nature.

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